Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *AFRIDOC_MT* | ***Context:*** *Ethiopia Ministry of Health — Amharic Clinical Translation Deployment*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (the reference translations and human evaluation labels themselves) raises the most consequential validity concerns. The four named Amharic translators [Q109] are recruited via a language coordinator [Q23, Q24] and attended a domain workshop with short translation guidelines [Q26], but the benchmark documents no MOH credentials, clinical backgrounds, EFDA training, or institutional affiliations — the deployment explicitly designates MOH-trained clinical translators as primary ground-truth authority and Amharic-speaking physicians as fallback. Web search did not find public credentials for these translators or a public MOH clinical translation certification registry [WEB-3 register variation; benchmark_validity_gaps_summary annotator_credentials NOT FOUND]. Inter-annotator agreement for Amharic human DA is α=0.46 [Q211, Q71], the lowest of the five languages, indicating noisy reference quality signal even within the benchmark's own framework. The translators' regional affiliation is undocumented, leaving regional Amharic register variation across Tigray/Oromia/Somali bureaus unaddressed. With OC marked HIGH priority and the user explicitly excluding general translators and academic linguists from authoritative status, the alignment is poor — score 2.

ID	Evidence
Q23	“We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language.” (p.3)
Q24	“The translators were recruited through a language coordinator who is also a native speaker of the language.” (p.3)
Q26	“Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1).” (p.3)
Q71	“We obtained Krippendorff's alpha values of ≥ 0.40, which are relatively low due to the fine granularity of the evaluation scale.” (p.8)
Q109	“Also, we also appreciate the translators whose names we have listed below: 1. Amharic: Bereket Tilahun, Hana M. Tamiru, Biniam Asmlash, Lidetewold Kebede 2. Hausa: Junaid Garba, Umma Abubakar, Jibrin Adamu, Ruqayya Nasir Iro 3. Swahili: Mohamed Mwinyimkuu, Laanyu koone, Baraka Karuli Mgasa, Said Athumani Said 4. Yorùbá: Ifeoluwa Akinsola, Simeon Onaolapo, Ganiyat Dasola Afolabi, Oluwatosin Koya 5. Zulu: Busisiwe Pakade, Rooweither Mabuya, Tholakele Zungu, Zanele Thembani” (p.11)
Q211	“We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively.” (p.26)
WEB-3	Amharic dialects mutually intelligible per Wikipedia, but health-register variation across bureaus is a separate question — unresolved (https://en.wikipedia.org/wiki/Amharic)

Strengths

- Four expert translators per language with a workshop on domain-specific terminology and short translation guidelines [Q23, Q26]
- Use of MT engines was prohibited and quality assurance described as capable of detection [Q120]
- AfriCOMET-based QE with manual review threshold (0.65) added a layer of automated quality control [Q27, Q28, Q29]
- Translators paid at clearly disclosed rates ($1,250 per 2,500 sentences, $55.15 per 80 documents for evaluators) — transparent compensation [Q31, Q198]
- Dataset analysis confirms specialized clinical terminology is rendered in Ethiopic script across complex topics [DATASET-D5, D43, D47]

ID	Evidence

Q23	“We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language.” (p.3)
Q26	“Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1).” (p.3)
Q27	“Quality control was conducted using automated quality estimation, followed by manual inspections by our language coordinators.” (p.3)
Q28	“We also used Quality Estimation (QE), specifically AfriCOMET (Wang et al., 2024a), Given a translated sentence in any African language and its corresponding source English sentence, AfriCOMET generates a score between 0 and 1, where 0 indicates poor quality and higher values signify better quality.” (p.3)
Q29	“The translators, in collaboration with the language coordinators, were tasked with reviewing instances that had quality estimation scores below 0.65.” (p.3)
Q31	“Each translator was paid $1, 250 for 2, 500 sentences.” (p.3)
Q120	“Use the language field to input the translations. It is essential not to rely on translation engines, as our quality assurance process can detect this. Depending on such tools may result in potential issues that you would need to address, leading to additional work on your part.” (p.18)
Q198	“Each annotator was remunerated with $55.15” (p.24)
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D43	ፀረ-ተህዋሲያን - ፀረ-ባክቴሪያ ፣ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርበግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels (reference translations and human DA scores) likely do NOT reflect the deployment's authoritative regional stakeholder population. The user designates MOH-trained clinical translators as primary and Amharic-speaking physicians as fallback ground-truth; the benchmark documents neither MOH credentials nor clinical/medical backgrounds for the four named Amharic translators [Q109]. No regional bureau affiliation is documented [register_variation web search NOT FOUND].

OC-2: Substantial disagreement potential between AFRIDOC-MT translators and MOH-credentialed clinical translators is plausible: the benchmark's translators may have used different terminology choices for AMR, drug names, infant feeding terms than MOH/EFDA standards prescribe [DATASET-D5, D43, D47, D44]. The α=0.46 Amharic agreement [Q211] further indicates that even the benchmark's own annotation pool produced noisy quality judgments — the lowest among five languages.

OC-3: Annotator demographics in the benchmark are minimally documented: names of four translators per language [Q109], recruited through a 'language coordinator who is also a native speaker' [Q24]. No professional credentials, institutional affiliations, regions, ages, or clinical backgrounds are disclosed. No formal datasheet/data statement detailing demographics is provided. This is a clear documentation gap relative to standard practice.

OC-4: Re-annotation by an MOH-credentialed clinical translator pool would be high-value: producing parallel reference translations or post-editing AFRIDOC-MT references to MOH/EFDA standards would yield a deployment-aligned reference set. Rough cost estimate scales from disclosed per-sentence rates [Q31].

OC-5: Aggregation methods: the benchmark uses single reference translations per row (one of four translators), and human DA averages three annotators per language [Q69, Q194]. Krippendorff's alpha is reported but no explicit handling of minority-perspective preservation is described. Given α=0.46 for Amharic, averaging may erase substantial annotator-level disagreement.

OC-6: Convergent validity is at risk: AFRIDOC-MT references may not converge with MOH-credentialed translator judgments [DATASET CRITICAL Concern 1]. External validity is at risk: high system scores against AFRIDOC-MT references do not reliably predict MOH-acceptable performance. Construct validity for the regional deployment is materially weakened by both annotator-credentials uncertainty and low Amharic IAA [Q211, DATASET CRITICAL Concern 4].

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Q23	“We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language.” (p.3)
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Q26	“Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1).” (p.3)
Q31	“Each translator was paid $1, 250 for 2, 500 sentences.” (p.3)
Q69	“In Table 10 we report average direct assessment (DA) scores (on a scale from 0 to 100) from three annotators per language for the health domain, when translating into four African languages.” (p.7)
Q109	“Also, we also appreciate the translators whose names we have listed below: 1. Amharic: Bereket Tilahun, Hana M. Tamiru, Biniam Asmlash, Lidetewold Kebede 2. Hausa: Junaid Garba, Umma Abubakar, Jibrin Adamu, Ruqayya Nasir Iro 3. Swahili: Mohamed Mwinyimkuu, Laanyu koone, Baraka Karuli Mgasa, Said Athumani Said 4. Yorùbá: Ifeoluwa Akinsola, Simeon Onaolapo, Ganiyat Dasola Afolabi, Oluwatosin Koya 5. Zulu: Busisiwe Pakade, Rooweither Mabuya, Tholakele Zungu, Zanele Thembani” (p.11)
Q194	“For the human evaluation, three native speakers of the African languages—primarily translators involved in the dataset creation—were recruited.” (p.24)
Q211	“We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively.” (p.26)
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D43	ፀረ-ተህዋሲያን - ፀረ-ባክቴሪያ ፣ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርበግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጨ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)

Information Gaps

- Professional credentials, MOH affiliation, clinical training, and regional location of the four named Amharic translators
- Whether the human DA evaluators (three native speakers per language, primarily drawn from the translation team [Q194]) hold any clinical credentials
- Whether the planned terminology harmonization [Q121] used any MOH/EFDA-aligned source
- Why Amharic IAA (α=0.46) is so much lower than Yorùbá (0.81) — could reflect translation ambiguity, register variation, or fine granularity of the scale

Requires Expert Verification

- Direct inquiry to AFRIDOC-MT authors (e.g., Israel Abebe Azime, Saarland University) regarding Amharic translator credentials and regional affiliations
- MOH clinical-translator side-by-side review of a sample of AFRIDOC-MT Amharic references to assess alignment with MOH standards
- Stakeholder elicitation from regional health bureau staff (Tigray, Oromia, Somali) on whether AFRIDOC-MT Amharic register is acceptable

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Remediation

Gap 1: AFRIDOC-MT Amharic translators have no documented MOH credentials or clinical backgrounds, and α=0.46 Amharic IAA indicates noisy reference signal — a direct mismatch with the deployment's MOH-trained-translator ground-truth authority.

Recommendation: Recruit 2–3 MOH-credentialed clinical translators (or Amharic-speaking physicians as fallback) to post-edit a sampled subset (e.g., 50–100 documents) of AFRIDOC-MT Amharic references to MOH/EFDA standards, producing a deployment-aligned evaluation set. Document each post-editor's MOH affiliation, clinical credentials, and regional bureau experience.

Gap 2: Regional Amharic register variation across Tigray, Oromia, Amhara, and Somali bureaus is unaddressed; benchmark uses single Addis Ababa-centric references whose regional acceptance is unverified.

Recommendation: Conduct stakeholder elicitation with regional health bureau staff in at least Oromia, Tigray, and Somali regions to identify any region-specific terminology preferences that would diverge from Addis Ababa-centric references. Document any findings as a register-variation flag set for evaluation.